S05 — Is temporal selection of a second cued letter suppressed, delayed and diffused with different time courses?

Question

In a rapid letter stream with two cued letters, does the report of the second cued letter change in efficacy (how often a report comes from around its position), latency (where reports are centred) and precision (their spread), with time courses across the 1-10 item interval that differ between measures? Predicted: a significant measure (3) x target (2) x SOA (10) interaction.

Data and sample

One file (`d04.csv`): 720 per-participant summary values = 12 participants x 3 measures x 2 targets x 10 SOAs. No missing values; every cell has 12 values. Practice trials were already discarded; **no exclusions**, n = 12 participants, 720 values analysed. Column mapping per the card: column `efficacy` = measure, `latency` = target, `precision` = SOA.

Test (named in advance)

Full-factorial fixed-effects ANOVA, all factors categorical, every value treated as independent (participant not in the model; equivalent to Matlab `anovan(..., 'model','full')`), Type III sums of squares (balanced design). Term of interest: measure x target x SOA. The claim holds if p < .05.

Result

Three-way interaction: F(18, 660) = 12.98, p = 3.3 x 10^-33, partial eta-squared = .26, 95% CI [.19, .30].

Cell means (averaged over participants):

measure	target	SOA 1	2	3	4	5	6	7	8	9	10
efficacy	T1	.124	.126	.128	.124	.124	.125	.125	.127	.127	.124
efficacy	T2	.098	.097	.084	.078	.076	.084	.096	.099	.110	.113
latency	T1	.24	.21	.10	.08	.10	-.02	.08	.06	.08	.02
latency	T2	-.22	-.62	-.82	-.14	.44	1.13	.92	.66	.68	.52
precision	T1	1.42	.91	1.30	1.27	1.18	1.11	1.24	1.07	1.12	1.06
precision	T2	1.75	1.98	3.25	4.25	2.17	1.27	1.40	1.45	1.18	1.38

For T1 all three measures are nearly flat across SOA. For T2, efficacy dips at SOA 3-5 and recovers by 9-10; variance (diffusion) peaks at SOA 3-4; latency is shifted *earlier* than the target at SOA 1-3 and *later* at SOA 5-10, peaking at SOA 6.

Checks (not primary)

- Measures are on different scales (probability, positions, positions squared), so part of an interaction across measures could be scale alone. With values z-scored within measure, the three-way term is still F(18, 660) = 10.18, p = 1.6 x 10^-25.
- Repeated-measures ANOVA with participant as a random factor (R `aov`, Error strata): three-way F(18, 198) = 16.87, p < 2 x 10^-16 (z-scored: F = 14.75).
- Target x SOA within each measure (fixed-effects): efficacy F(9, 220) = 1.98, p = .043; latency F(9, 220) = 12.57, p < 10^-15; precision F(9, 220) = 8.96, p < 10^-10. The efficacy effect is small in absolute terms (a dip of ~.05 against ~.12).

Caveats

Treating 720 values from 12 people as independent inflates the degrees of freedom; the card specifies that test, and the participant-aware ANOVA gives the same answer. The early-SOA latency shift for T2 is negative (reports before the target), not "delayed"; the delay appears from SOA 5 on. Efficacy values are per-position probabilities (~1/8), so its interaction is the weakest of the three.

Conclusion

The claim holds on this data: the measure x target x SOA interaction is significant and large (F(18, 660) = 12.98, p < .001, partial eta-squared = .26), and survives rescaling and a participant-level ANOVA.

Files: `analysis.py` (all numbers), `analysis.out`, `log.md`.